# COMPX551 Assignment 4

Contribution Statement — Group 20

## Overview

We split the work four ways, one workload per person, each matching a stage of the app's pipeline: sensing, signal processing, notifications, and UI. Our Android app reads EEG from a Muse 2 headset, estimates whether the user is focused, holds back notifications while they are, and delivers one catch-up summary afterwards.

## Jesse Du Plessis: Sensing and service

Jesse connected the app to the headset and built the background service that runs the whole decision loop. Early on, he provided a fake EEG source and an OSC bridge to the Mind Monitor app, so the team had data to build against before the Bluetooth connection worked. He also wrote the Architecture and device integration slides.

**Files:** `MuseEegSource.kt` (Bluetooth connection to the Muse 2 through LibMuse, one epoch per second), `OscEegSource.kt` (EEG from Mind Monitor over OSC), `FakeEegSource.kt` (simulated EEG for testing), and `FocusService.kt` (the background service running the decision loop, with reconnect and headset-removed handling).

## Hamish Gilmore: Signal processing and calibration

Hamish built the processing that turns EEG into a focus state, and the calibration that fits it to each user. He also wrote the Data Processing and Cultural and Ethical Considerations slides.

**Files:** `FocusPipeline.kt` (the focus engine), `ArtifactRejector.kt` (removes blinks, poor contact and muscle noise), `FocusStateMachine.kt` (two-threshold focus decision), `DriftDetector.kt` (recalibration prompt), `CalibrationSession.kt` and `ThresholdFitter.kt` (fit per-user thresholds and AUC), `ThresholdStore.kt` (saves thresholds), `ForegroundEegSource.kt` and the Muse connection fixes, plus nine unit test files.

## Mohammad Emon: Notifications

Mohammad built notification suppression: turning on Do Not Disturb while the user is focused, collecting the notifications that arrive, and releasing them as one summary afterwards. He also wrote the Application and its Purpose slide.

**Files:** `NotificationController.kt` (switches suppression with the focus state and flushes the summary), `DndController.kt` (the Do Not Disturb rule), `FocusNotificationListener.kt` (captures incoming notifications), `DeferralQueue.kt` (holds them during focus) and `BreakthroughPolicy.kt`.

## Richard Larsen: UI, onboarding and stats

Richard built every screen in the app, plus the calibration task screens and the session history behind the stats. He also set up the project scaffold, drafted the shared models and interfaces, merged the workload branches, created the slide template, and wrote the Visualisation and Historical/Summary Data and Reflection and Future Work slides.

**Files:** `MainActivity.kt` and `FocusNavHost.kt` (entry point and navigation), the Home, Stats, Settings, Onboarding and Calibration screens, the four calibration task screens, shared charts and components, the theme, `SessionStore.kt` (session history and stats), `UiPreferences.kt`, and UI unit tests.

**All members have read this statement and agree it reflects their contributions.**